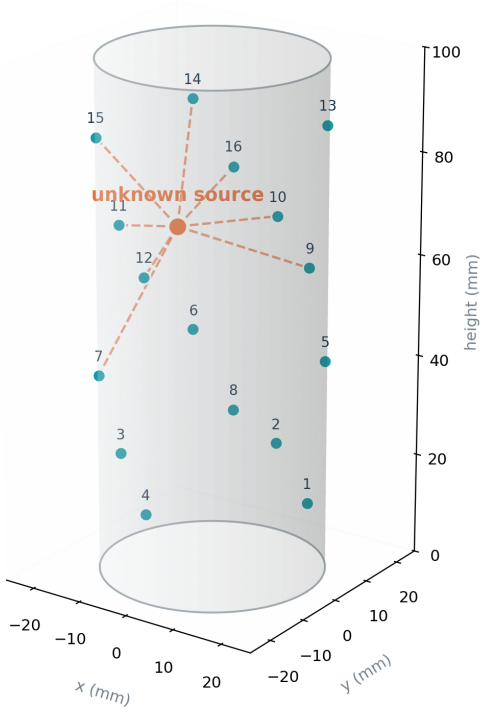
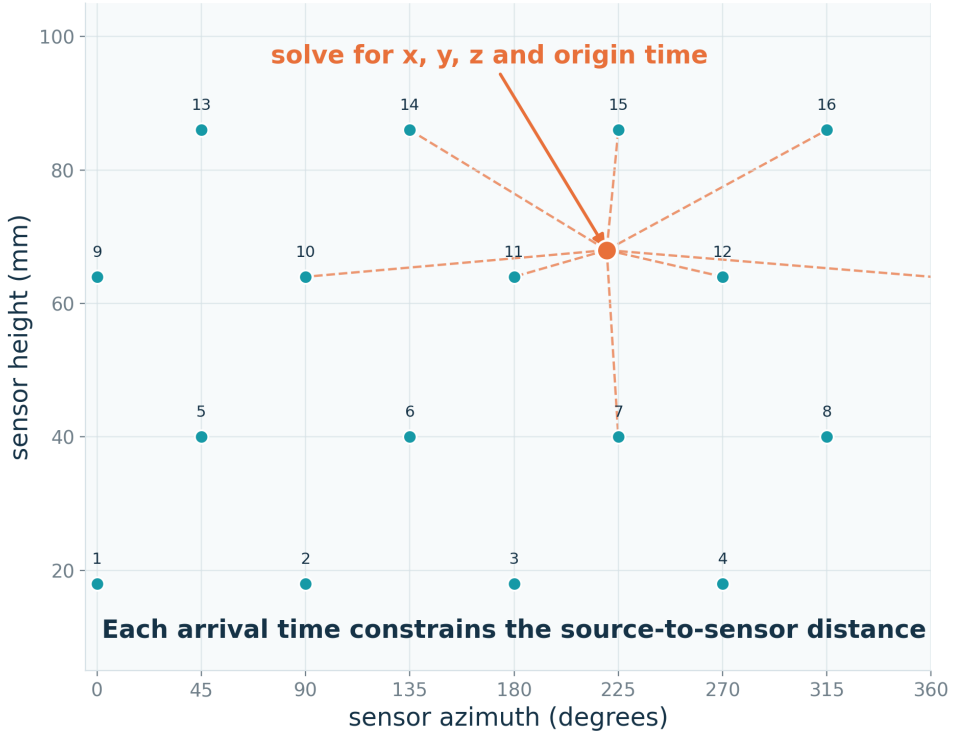
Arrival times turn an array into a spatial measurement

3-D sensor array



Cylinder unwrapped into an angle-height map



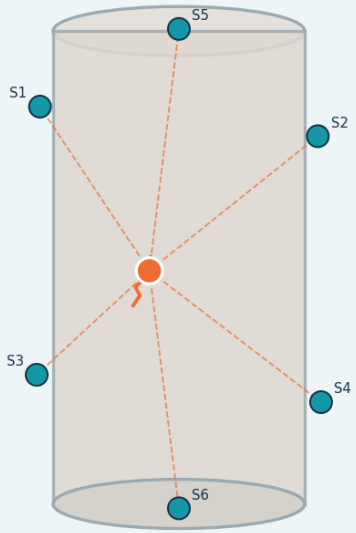
- Each arrival defines a distance–time constraint.
- Several sensors solve for origin time and 3-D position.
- Residuals show how well the assumed velocity model explains the picks.

More sensors improve geometry; they cannot rescue a systematically wrong VP model.

Activity 3

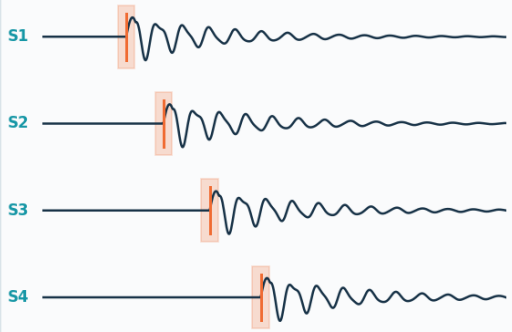
A precise dot can still be in the wrong place
Location uncertainty

1 ARRIVALS



Each sensor supplies an arrival-time constraint

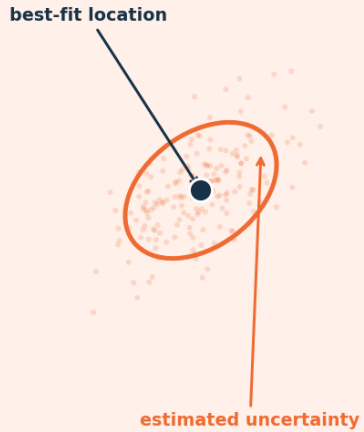
2 PICK + MODEL



pick uncertainty

+ assumed V_P field
+ sensor geometry

3 LOCATION



A small ellipsoid can still be biased if the velocity model is wrong.

GEOMETRY

events outside the array are weakly constrained

PICKING

noisy or emergent arrival

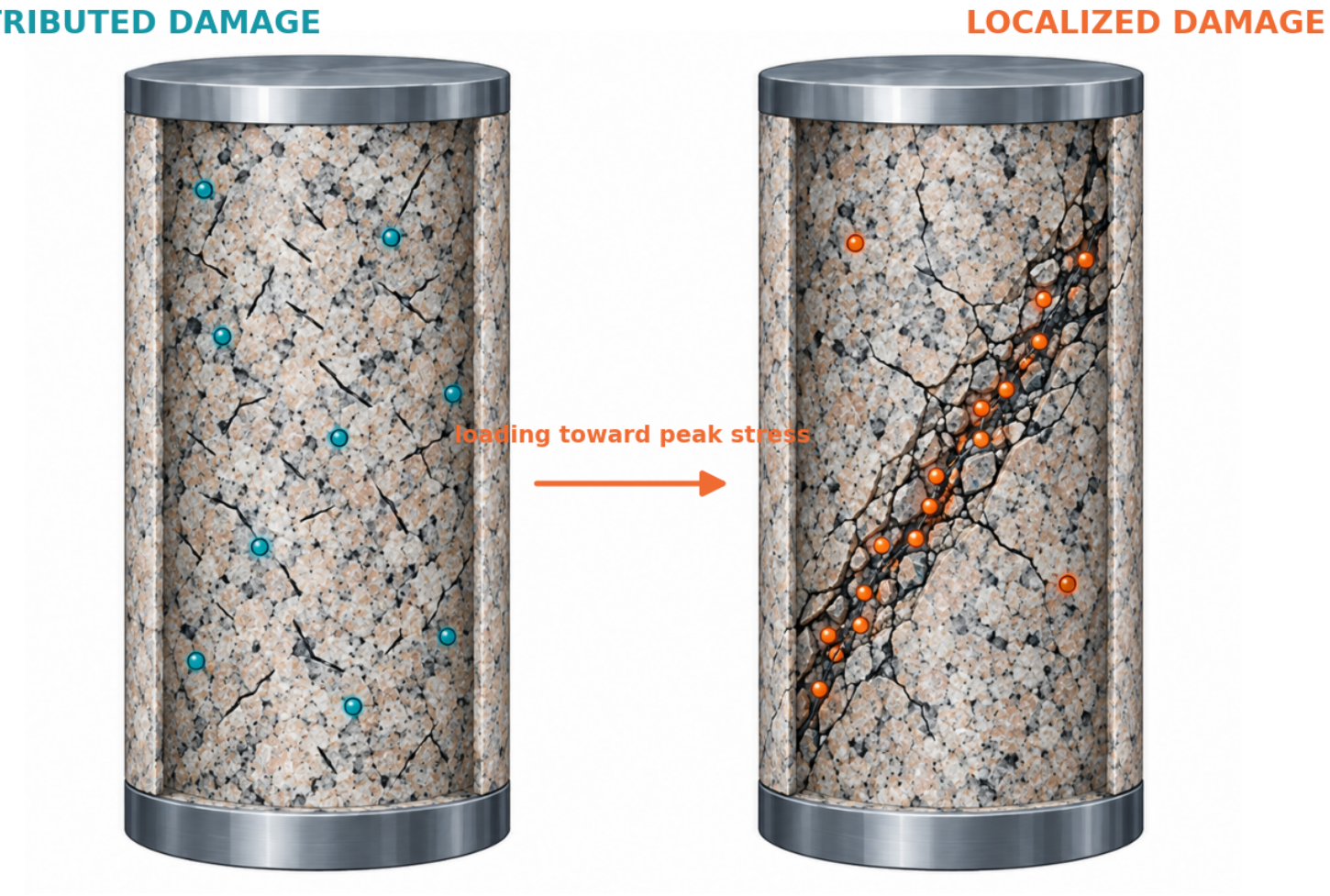
VELOCITY

VP evolves and becomes anisotropic

PATH

attenuation and scattering hide first motion

When should a cloud of AE events become a fault?

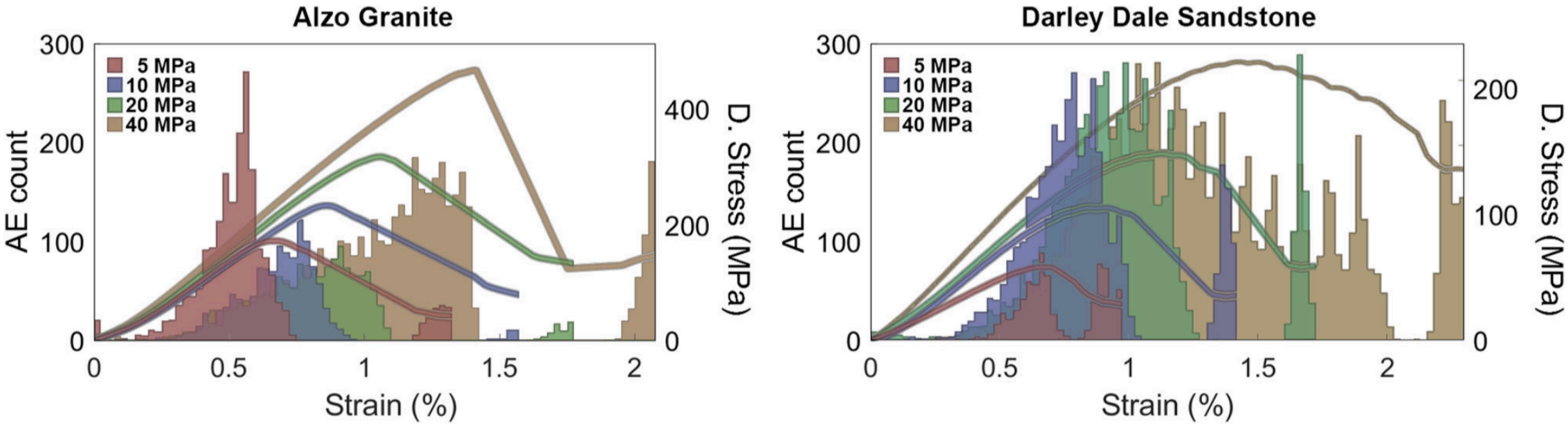


sparse microcracks and broadly scattered AE

coalesced fracture band and clustered AE

Localization is a spatial reorganization of damage, not merely an increase in event count.

Granite vs sandstone: where do different cracks form?



AE source mechanisms evolve in space through nucleation, coalescence and failure.

King et al. (2021), JGR Solid Earth, Figures 5–6 · DOI: 10.1029/2020JB021059